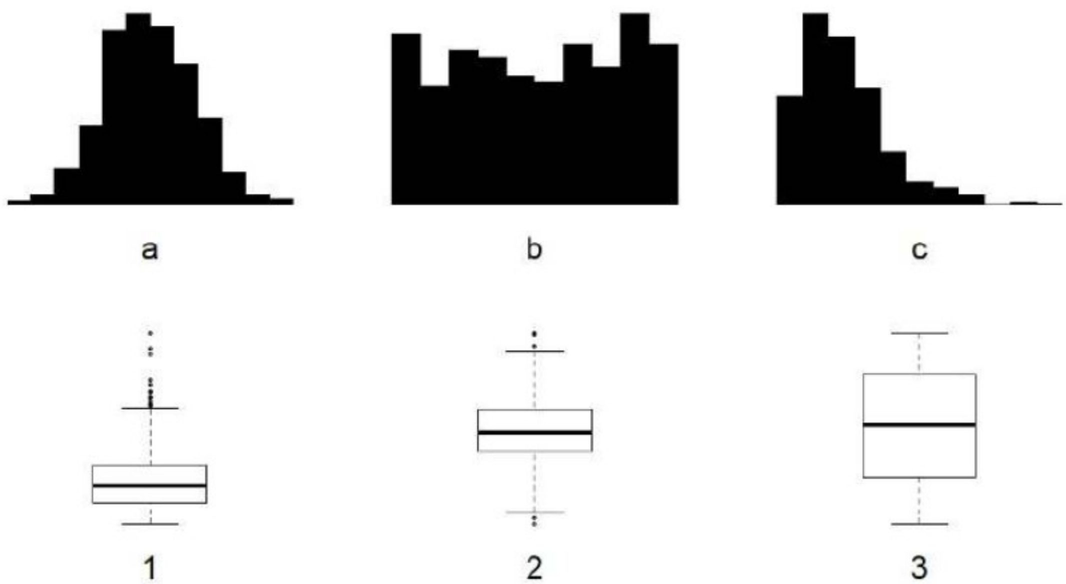
Problem1

A researcher would like to study the effect of eating breakfast on a cognitive function. Volunteers are recruited throughthe study by posting flyers on campus. He randomly assigns subjects to two groups, one told to eat breakfast beforeparticipating in the cognitive test and one asked to eat breakfast after the test. However, he suspects that people's usualbreakfast habits (whether or not they usually eat breakfast at all) may influence his results. In order to address this, whatshould he do prior to assigning subjects to experimental groups?

(1 point)

Problem 2

The figure below displays the histogram and boxplot of three different datasets. Match the histogram and boxplot,which belong to the same dataset. You do not need to justify your choice.



(2 points)

Problem 3

Consider the following distribution of car ownership and public transportation usage.

```
##                    pub_trans
## car                 does not use pub. transport uses pub. transport Sum
##   does not own car                           20               100 120
##   owns car                                  126                34 160
##   Sum                                       146               134 280
```

Is owning a car independent of using public transport? Justify your answer. (3 points)

Problem 4

When a variable follows a normal distribution, what percentage of observations are contained within 1.3 standard deviationsof the mean? Justify your answer

Hint

```
pnorm(1.3, lower.tail = FALSE)
```

```
## [1] 0.09680048
```

(3 points)

Problem 5

For a t-test in R, we obtained the following output.

```
t = 0.2024, df = 14, p-value = 0.8425
alternative hypothesis: true mean is not equal to 20
95 percent confidence interval:
18.91483 21.31131
sample estimates:
mean of x
20.11307
```

Decide if the following statements are true or false. Justify your **answer**.

a The margin of error of the computed confidence interval is equal to 1.19824.

b The following R-code pt(0.2024, $df = 14$) will result in an output of 0.8425.

c The null hypothesis tested here is H_0: $\mu = 20.11307$.

d $[18.97138, 21.25476]$ is a 99% confidence interval for the true mean.

Remark Only correct justifications are awarded with points. Not the decision (true/false) (4 points)

Problem 6

Suppose a paired t-test yields a p-value of 0.03 when testing hypotheses H_0: $\mu_d = 0$ vs H_A: $\mu_d \neq 0$. Decide if $(1.3, 3.4)$ or $(-1.3, 3.4)$ is the corresponding 95% confidence interval for the difference in means. Justify your **answer**. (2 points)

Problem 7

Subjects from Central Prison in Raleigh, NC, volunteered for an experiment involving an "isolation" experience. The goalof the experiment was to find a treatment that reduces subjects' psychopathic deviant T scores. This score measures aperson's need for control or their rebellion against control. 42 subjects were randomly assigned to three treatment groups.

(6 points)

```
prison %>%
  group_by(treatment) %>%
  summarise(mean(score), sd(score), n())
```

```
## # A tibble: 3 x 4
##   treatment `mean(score)` `sd(score)` `n()`
##   <chr>             <dbl>       <dbl> <int>
## 1 Trt1               6.21       12.3     14
## 2 Trt2               2.86        7.94    14
## 3 Trt3              -3.21        8.57    14
```

The success of the different treatments is analysed with an ANOVA test:

```
## Analysis of Variance Table
##
## Response: score
##           Df Sum Sq Mean Sq F value  Pr(>F)
## treatment  2  639.5  319.74  3.3338 0.04607 *
## Residuals 39 3740.4   95.91
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

a) State the null hypothesis of the ANOVA test.

b) What is the conclusion of the test? Use a 5% significance level. Justify your **answer**.

c) Use

```
TukeyHSD(aov(score ~ treatment, data = prison))
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = score ~ treatment, data = prison)
##
## $treatment
##                diff       lwr       upr     p adj
## Trt2-Trt1 -3.357143 -12.37517  5.660884 0.6391994
## Trt3-Trt1 -9.428571 -18.44660 -0.410545 0.0387614
## Trt3-Trt2 -6.071429 -15.08946  2.946598 0.2411800
```

to determine which groups are different from each other at the (global) significance level $\alpha = 0.05$. Justify your **answer**.

Problem 8

An experiment was conducted to investigate the effectiveness of a certain medicine by comparing outcomes from a treatmentand a control group. 16 of the 20 patients who take the medicine improved their condition, while 45 of the 78 in the controlgroup improved. What is the margin of error to estimate the difference in the proportion of subjects that experiencedimprovement with a 95% confidence interval?

Hint:

```
qnorm(c(0.05, 0.95, 0.975))
```

```
## [1] -1.64  1.64  1.96
```

(3 points)

Problem 9

The World Bank reports that 1.7% of the US population lives on less than $2 per day. A policy maker claims that thisnumber is misleading because of variation from state to state and rural to urban. To investigate this, she takes a randomsample of 100 individuals in Atlanta to compare with the national average and finds that 4% of the Atlanta populationlive on less than $2 per day.

a State the alternative hypothesis to test whether Atlanta has a significantly higher percentage than the nationalaverage.

b Test whether Atlanta has a significantly higher percentage than the national average at the significance level $\alpha = 0.05$

```
pbinom(0:5, 100, prob = 0.017)
```

```
## [1] 0.180 0.491 0.758 0.908 0.972 0.993
```

(4 points)

Problem 10

A study is designed to test whether there is a relationship between natural hair colour ($brunette, blond, red$) and eyecolour ($blue, green, brown$). A large χ^2 test statistic value is obtained.

What conclusions can be drawn regarding the relationship between natural hair and eye colour? (1 point)

Problem 11

76 Starbucks food items were analysed for their calorie and carbohydrate content. We used linear regression to explorethe relationship between the number of calories and the amount of carbohydrates (in grams) contained in Starbucks foodmenu items. We obtained the following model (3 points)

```
model
```

```
##
## Call:
## lm(formula = carb ~ calories, data = starbucks)
##
## Coefficients:
## (Intercept)     calories
##       8.944        0.106
```

which has a R^2 value of 0.46.

a) Give an interpretation of the intercept.

b) Give an interpretation of the slope.

c) What is the correlation between calories and carbohydrates?

Problem 12

The common brush tail possum of the Australia region is a bit cuter than its distant cousin, the American opossum. Weconsider 104 brush tail possums from two regions in Australia, where the possums may be considered a random samplefrom the population. The first region is Victoria, which is in the eastern half of Australia and traverses the southern coast.The second region consists of New South Wales and Queensland, which make up eastern and north-eastern Australia. Weuse logistic regression to differentiate between possums in these two regions.

The outcome variable, called pop, takes value Vic when a possum is from Victoria and other (reference level) when it isfrom New South Wales or Queensland. We consider four predictors: sex,totalL (total length) and tailL (tail length).
A summary of the fitted model is the following:

```
## # A tibble: 4 x 5
##   term        estimate std.error statistic     p.value
##   <chr>          <dbl>     <dbl>     <dbl>       <dbl>
## 1 (Intercept)   22.2      7.51        2.95 0.00316
## 2 sexm          -1.50     0.617      -2.43 0.0152
## 3 totalL         0.407    0.0967      4.21 0.0000254
## 4 tailL         -1.55     0.298      -5.19 0.000000216
```

a) Write out the form of the fitted model.

b) Give an interpretation of the estimated slope of `sex` in terms of the odds ratio.

(4 points)